CHEM 5260 Midterm Project
A preliminary theoretical and numerical investigation on quantum relaxation

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(Dated: March 24, 2026)

We present a combined analytical and numerical study of quantum relaxation dynamics in two finite-dimensional doorway-state models. Model A consists of a single doorway state $|1\rangle$ coupled to a discrete manifold of bath states $\{|k\rangle\}$, while Model B adds an upstream state $|0\rangle$ so that population flows sequentially as $|0\rangle \to |1\rangle \to \{|k\rangle\}$. The analytical treatment is based on Green's functions and self-energies under the wide-band approximation, while the exact finite-dimensional dynamics is obtained independently from the full Hermitian Hamiltonian by exact diagonalization. For Model A the theory recovers the Fermi's-golden-rule prediction $P_1(t) = e^{-\Gamma t}$ with $\Gamma = 2\pi|V|^2\rho$, and the numerics show that this exponential regime is realized when $\rho^{-1} \lesssim \Gamma \ll \Delta E$, while finite bath size, narrow bandwidth, or strong coupling produce non-Markovian oscillations and Poincaré recurrences. For Model B, the same wide-band reduction assumes that the doorway state $|1\rangle$ undergoes Markovian exponential decay with rate Γ, so that the pair $\{|0\rangle, |1\rangle\}$ becomes a damped two-level system with overdamped, critically damped, and underdamped regimes separated by the threshold $V_{01} = \Gamma/4$. Numerical scans over N, ΔE, V_{1k}, V_{01}, and the detuning $E_0 - E_1$ quantitatively confirm the analytical predictions, including the Lorentzian resonance profile and the crossover from monotonic transfer to damped Rabi oscillations. Random-sign couplings are found to be gauge-equivalent to uniform couplings, whereas Gaussian-distributed couplings preserve the mean decay trend but broaden the ensemble of trajectories.

Keywords: quantum relaxation, doorway-state model, Fermi's golden rule, wide-band approximation, damped two-level system, Poincaré recurrence

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I. INTRODUCTION

Quantum relaxation is a fundamental phenomenon in chemical physics that underlies processes ranging from intramolecular vibrational energy redistribution and electronic energy transfer to non-radiative decay and photochemical reactions [1]. In a minimal doorway-state picture, an initially populated "bright" state is coupled to a dense manifold of background states, and population leaks out on a timescale governed by the manifold density of states and the coupling matrix elements.

The analytical theory of such processes, developed via Green's function and self-energy methods [1, 2], predicts that the survival probability of the doorway state decays exponentially in time when the manifold is sufficiently broad and dense to be treated as a quasi-continuum. The exponential decay rate is given by Fermi's golden rule (FGR):

$$\Gamma = 2\pi|V|^2\rho, \tag{1}$$

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where ρ is the density of states of the bath manifold and V is a characteristic coupling matrix element; throughout this paper we work in units where $\hbar = 1$. This result rests on the *wide-band approximation*: the resonance width Γ must be small relative to the total bandwidth ΔE of the manifold, while remaining large compared to the discrete level spacing ρ^{-1}:

$$\rho^{-1} \lesssim \Gamma \ll \Delta E. \tag{2}$$

When these inequalities are violated the dynamics can deviate markedly from exponential, exhibiting quantum revivals, persistent oscillations, or incomplete relaxation. In other words, the irreversible Markovian decay predicted by FGR is not a generic property of a finite bath, but a specific limit that must be demonstrated.

In any finite numerical simulation, however, the manifold is discrete and bounded, so the continuum picture can only emerge as an approximation over a finite time window. Understanding when and how that approximation breaks down is therefore essential.

The present work addresses these questions through a combined analytical and numerical study of two finite-dimensional model Hamiltonians. Model A is the minimal doorway-state problem: a single state $|1\rangle$ coupled to a bath manifold $\{|k\rangle\}$. Model B extends this construction by adding an upstream state $|0\rangle$ coupled only to $|1\rangle$, so that the population flow proceeds sequentially: $|0\rangle \to |1\rangle \to \{|k\rangle\}$.

For Model A, Green's-function and self-energy methods [1, 2] recover the standard wide-band result $P_1(t) = e^{-\Gamma t}$. For Model B, the same wide-band approximation implies that the doorway state $|1\rangle$ decays Markovianly with rate Γ, so that the remaining pair $\{|0\rangle, |1\rangle\}$ forms an effective damped two-level system, i.e. a damped two-state Rabi problem [3]. This yields a clear analytical classification of the dynamics: overdamped for $V_{01} < \Gamma/4$, critical at $V_{01} = \Gamma/4$, and underdamped for $V_{01} > \Gamma/4$. Detuning $E_0 - E_1$ then enters as a resonance parameter that suppresses the upstream-to-doorway transfer through a Breit–Wigner / Lorentzian line shape.

The numerical part of the work is based on exact diagonalization of the full finite Hamiltonian, with no time-discretization or perturbative approximation. By varying N, ΔE, V_{1k}, V_{01}, and $E_0 - E_1$, we test the analytical predictions against the exact dynamics, identify the boundaries of the wide-band regime, and compare uniform, random-sign, and Gaussian bath couplings.

The central questions addressed here are: (i) under what conditions does a finite discrete bath reproduce the exponential relaxation predicted by Fermi's golden rule? (ii) how does breakdown of the wide-band approximation manifest in finite systems? and (iii) how does the addition of an upstream state generate distinct overdamped, critical, and underdamped regimes?

II. MODEL HAMILTONIANS AND METHODS

A. Model Hamiltonian

1. Model A

The Hilbert space of Model A is spanned by the orthonormal basis $\{|1\rangle, |k_1\rangle, |k_2\rangle, \ldots, |k_N\rangle\}$. The full Hamiltonian is

$$\hat{H}_A = E_1 |1\rangle\langle 1| + \sum_{k=1}^{N} E_k |k\rangle\langle k| + \sum_{k=1}^{N} \big(V_{1k} |1\rangle\langle k| + V_{k1} |k\rangle\langle 1|\big), \tag{3}$$

where E_1 is the energy of the doorway state and E_k are the energies of the bath levels. All couplings are taken real so that $V_{k1} = V_{1k}$. The bath energies are distributed uniformly over an interval of width ΔE centred on E_1,

$$E_k \in \Big[E_1 - \tfrac{\Delta E}{2},\ E_1 + \tfrac{\Delta E}{2}\Big], \tag{4}$$

giving a continuum density of states $\rho = N/\Delta E$. For the uniformly spaced numerical grids used below, the exact nearest-neighbour spacing is $\Delta\omega = \Delta E/(N-1)$, so that in the large-$N$ limit one has $\rho^{-1} \approx \Delta E/N \approx \Delta\omega$. Without loss of generality we set $E_1 = 0$.

The system is initialized in the doorway state, $|\psi(0)\rangle = |1\rangle$, and the principal observable is the survival probability

$$P_1(t) = |\langle 1|\psi(t)\rangle|^2. \tag{5}$$

We also monitor the total bath population $P_K(t) = \sum_{k=1}^{N} |\langle k|\psi(t)\rangle|^2 = 1 - P_1(t)$.

2. Model B

Model B introduces an additional state $|0\rangle$ that is coupled only to the doorway state $|1\rangle$, with no direct coupling to the bath. The Hilbert space is thus $\{|0\rangle, |1\rangle, |k_1\rangle, \ldots, |k_N\rangle\}$ and the Hamiltonian is

$$\begin{aligned}\hat{H}_B = {} & E_0 |0\rangle\langle 0| + E_1 |1\rangle\langle 1| + V_{01} |0\rangle\langle 1| + V_{10} |1\rangle\langle 0| \\ & + \sum_{k=1}^{N} E_k |k\rangle\langle k| + \sum_{k=1}^{N} \big(V_{1k} |1\rangle\langle k| + V_{k1} |k\rangle\langle 1|\big).\end{aligned} \tag{6}$$

The total number of levels is $N + 2$ and the initial condition is $|\psi(0)\rangle = |0\rangle$. The observables are

$$P_0(t) = |\langle 0|\psi(t)\rangle|^2, \tag{7}$$

$$P_1(t) = |\langle 1|\psi(t)\rangle|^2, \tag{8}$$

$$P_K(t) = \sum_{k=1}^{N} |\langle k|\psi(t)\rangle|^2. \tag{9}$$

The energy-level structures of both models are illustrated in Fig. 1.

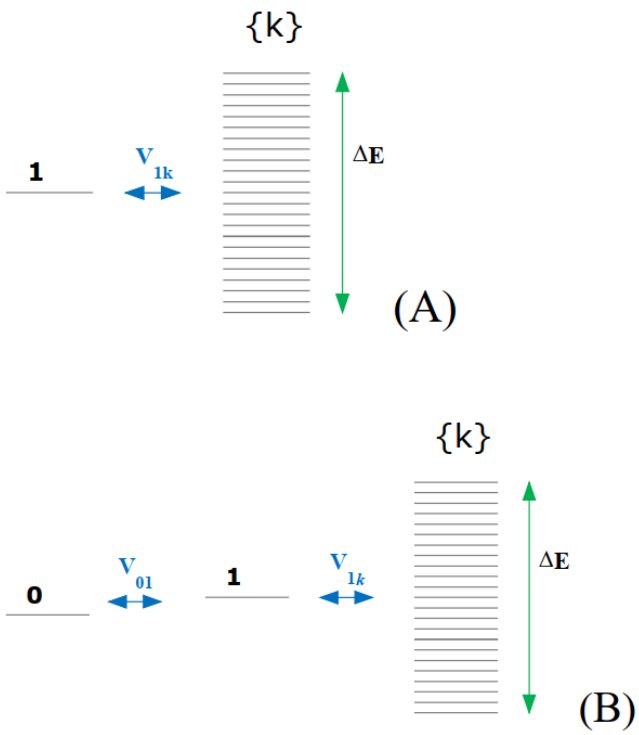

FIG. 1. Energy-level diagrams for (A) Model A, in which the doorway state $|1\rangle$ is coupled directly to the bath manifold $\{|k\rangle\}$ with matrix elements V_{1k}; and (B) Model B, in which the upstream state $|0\rangle$ is first coupled to $|1\rangle$ via V_{01}, and $|1\rangle$ subsequently couples to the manifold via V_{1k}.

3. *Coupling schemes*

Three coupling schemes are compared throughout.

1. *Uniform couplings*: all matrix elements equal, $V_{1k} = V$ for $k = 1, \ldots, N$.
2. *Random-sign couplings*: $V_{1k} = \pm V$ with equal probability, so that $\langle V_{1k}\rangle = 0$ and $\langle V_{1k}^2\rangle^{1/2} = V$.
3. *Gaussian couplings*: $V_{1k} \overset{\text{i.i.d.}}{\sim} \mathcal{N}(0, V^2)$, so that $\langle V_{1k}\rangle = 0$ and $\langle V_{1k}^2\rangle^{1/2} = V$.

Throughout, we denote the root-mean-square doorway–bath coupling by

$$V_{\text{rms}} \equiv \langle V_{1k}^2\rangle^{1/2},$$

so that $V_{\text{rms}} = V$ for all three schemes above. Whenever a single characteristic coupling scale is needed analytically, we write $V \equiv V_{\text{rms}}$.

B. Time-dependent Schrödinger equation and exact diagonalization

Since both Hamiltonians are represented by finite Hermitian matrices of dimension $(N+1)\times(N+1)$ (Model A) or $(N+2)\times(N+2)$ (Model B), the dynamics is governed by the time-dependent Schrödinger equation (TDSE),

$$i\hbar \frac{d}{dt}\,|\psi(t)\rangle = \hat{H}\,|\psi(t)\rangle\,, \tag{10}$$

whose formal solution is $|\psi(t)\rangle = e^{-i\hat{H}t/\hbar}\,|\psi(0)\rangle$. Expanding in the energy eigenbasis $\hat{H}\,|j\rangle = E_j\,|j\rangle$,

$$|\psi(t)\rangle = \sum_j e^{-iE_j t}\,|j\rangle\,\langle j|\psi(0)\rangle\,, \tag{11}$$

where $\{E_j, |j\rangle\}$ are the eigenvalues and eigenvectors of H. Numerically this spectral decomposition is obtained via the full diagonalization $H = UDU^\dagger$, which gives $e^{-iHt} = Ue^{-iDt}U^\dagger$. This approach is exact within machine precision and avoids all time-stepping errors. We work in units where $\hbar = 1$ throughout, and all parameters are reported in consistent dimensionless units. Typical system sizes range from $N = 50$ to $N = 200$ bath levels; convergence with respect to N is checked explicitly.

C. Green's function treatment and the wide-band approximation

The analytical benchmark for both models is obtained by combining the retarded Green's function with the wide-band approximation [1, 2]. For Model A this leads directly to Fermi's-golden-rule exponential decay. For Model B, after the same wide-band reduction of the bath, the remaining dynamics is that of a damped two-level system formed by the pair $\{|0\rangle, |1\rangle\}$.

1. *Model A: Green's function treatment under the wide-band approximation*

Equations of motion for the amplitudes. Projecting the TDSE onto the basis states of Model A and writing $c_1(t) = \langle 1|\psi(t)\rangle$ and $c_k(t) = \langle k|\psi(t)\rangle$, the coupled amplitude equations are ($\hbar = 1$)

$$i\dot{c}_1 = E_1 c_1 + \sum_k V_{1k}\, c_k, \tag{12}$$

$$i\dot{c}_k = E_k c_k + V_{k1}\, c_1. \tag{13}$$

Formally integrating Eq. (13) with $c_k(0) = 0$ and substituting back into Eq. (12) yields an integro-differential equation for $c_1(t)$, whose Laplace transform is most naturally analysed via the Green's function.

Dyson equation and the self-energy. The retarded Green's operator $\hat{G}(E)$ is defined as

$$\hat{G}(E) = \lim_{\eta\to 0^+} (E - \hat{H} + i\eta)^{-1}, \tag{14}$$

and the survival amplitude is recovered by the inverse Fourier transform

$$c_1(t) = \int_{-\infty}^{\infty} \frac{dE}{2\pi i}\, e^{-iEt}\, G_{11}(E), \qquad G_{11}(E) \equiv \langle 1|\hat{G}(E)|1\rangle\,. \tag{15}$$

Projecting the resolvent identity $(E-\hat{H})\hat{G}=\hat{1}$ onto $|1\rangle$ and onto each $|k\rangle$ gives

$$G_{k1}(E)=\frac{V_{k1}}{E-E_k+i\eta}\,G_{11}(E), \tag{16}$$

and back-substituting yields the Dyson equation

$$G_{11}(E)=\frac{1}{E-E_1-\Sigma(E)}, \tag{17}$$

where the self-energy is

$$\Sigma(E)=\sum_{k=1}^{N}\frac{|V_{1k}|^2}{E-E_k+i\eta}. \tag{18}$$

Wide-band approximation and exponential decay. In the continuum limit Eq. (18) becomes an integral over the bath density of states $\rho(\varepsilon)$:

$$\Sigma(E)\to\int_{-\infty}^{\infty}d\varepsilon\,\frac{|V(\varepsilon)|^2\rho(\varepsilon)}{E-\varepsilon+i\eta}. \tag{19}$$

Applying the Sokhotski–Plemelj identity $\lim_{\eta\to0^+}(x+i\eta)^{-1}=\mathcal{P}(1/x)-i\pi\delta(x)$ decomposes $\Sigma=\Delta(E)-\frac{i}{2}\Gamma(E)$ with the energy-dependent decay rate

$$\Gamma(E)=2\pi|V(E)|^2\rho(E). \tag{20}$$

The *wide-band approximation* (WBA) assumes $|V(\varepsilon)|^2\rho(\varepsilon)\approx|V|^2\rho=$ const across the resonance width, which holds when $\Gamma\ll\Delta E$ and E_1 lies well away from the band edges. The energy shift Δ then vanishes by symmetry and the self-energy reduces to a pure imaginary constant,

$$\Sigma(E)\approx-\frac{i\Gamma}{2},\qquad\Gamma=2\pi|V|^2\rho. \tag{21}$$

The Green's function $G_{11}(E)$ (17) becomes a Lorentzian pole,

$$G_{11}(E)\approx\frac{1}{E-E_1+i\Gamma/2}, \tag{22}$$

of half-width $\Gamma/2$ centered at E_1. Evaluating the inverse transform (15) by closing the contour in the lower half-plane ($t>0$) and picking up the single pole at $E=E_1-i\Gamma/2$,

$$c_1(t)=e^{-iE_1t}\,e^{-\Gamma t/2}, \tag{23}$$

so that the survival probability decays exponentially,

$$P_1(t)=|c_1(t)|^2=e^{-\Gamma t},\qquad\Gamma=2\pi|V|^2\rho, \tag{24}$$

in agreement with Fermi's golden rule (1). The validity conditions are exactly those of Eq. (2); violations of either bound lead to deviations from exponential behavior explored numerically in the next section.

2. *Model B: damped two-level dynamics under the wide-band approximation*

For Model B the bath still couples only through the doorway state $|1\rangle$, so the same wide-band self-energy can be used after projecting onto the two-dimensional subspace $\{|0\rangle,|1\rangle\}$. The bath degrees of freedom are integrated out exactly as in Model A, yielding the effective non-Hermitian Hamiltonian

$$H_{\rm eff}^{(B)}=\begin{pmatrix}E_0 & V_{01}\\ V_{01} & E_1-i\Gamma/2\end{pmatrix},\qquad\Gamma=2\pi|V_{1k}|^2\rho. \tag{25}$$

The crucial assumption here is the *wide-band approximation*: the bath seen by $|1\rangle$ is taken to be sufficiently broad and dense that $\Sigma(E)\approx-i\Gamma/2$ is energy independent, so the isolated doorway state would decay Markovianly as $c_1(t)\propto e^{-iE_1t}e^{-\Gamma t/2}$. Only under this assumption can Model B be reduced to an ordinary damped two-level (Rabi) problem [3]. Equivalently, the amplitudes $c_0(t)$ and $c_1(t)$ obey

$$i\dot{c}_0=E_0c_0+V_{01}c_1, \tag{26}$$

$$i\dot{c}_1=V_{01}c_0+\left(E_1-i\frac{\Gamma}{2}\right)c_1. \tag{27}$$

The complex normal-mode energies are obtained from the characteristic equation of $H_{\rm eff}^{(B)}$:

$$E_\pm=\frac{E_0+E_1-i\Gamma/2}{2}\pm\frac{1}{2}\sqrt{(E_0-E_1+i\Gamma/2)^2+4V_{01}^2}, \tag{28}$$

which govern the exact WBA dynamics of the amplitudes $c_0(t)$ and $c_1(t)$. At finite detuning $\Delta_{01}=E_0-E_1$, these poles interpolate between coherent two-level oscillations and irreversible decay into the continuum. If the wide-band approximation fails, the decay of $|1\rangle$ is no longer strictly Markovian and the simple damping classification below ceases to be quantitatively valid.

The analytical treatment above relies on two linked assumptions of the wide-band approximation. First, the bath density of states and coupling strength must vary only slowly across the resonance width, so that the self-energy may be treated as effectively energy independent. Second, the bath must be sufficiently dense on the timescale of the decay that it can be treated as effectively continuous, i.e. $\rho^{-1}\lesssim\Gamma$. Violation of these assumptions through finite level spacing, narrow bandwidth, or strong coupling leads to the non-Markovian oscillations and Poincaré recurrences observed in the numerical results below.

The damping classification is most transparent on resonance, $E_0=E_1\equiv E_*$. Writing $c_j(t)=e^{-iE_*t}\tilde{c}_j(t)$ gives

$$i\dot{\tilde{c}}_0=V_{01}\tilde{c}_1, \tag{29}$$

$$i\dot{\tilde{c}}_1=V_{01}\tilde{c}_0-i\frac{\Gamma}{2}\tilde{c}_1. \tag{30}$$

Eliminating $\tilde{c}_1$ yields

$$\ddot{\tilde{c}}_0+\frac{\Gamma}{2}\dot{\tilde{c}}_0+V_{01}^2\tilde{c}_0=0, \tag{31}$$

which is mathematically identical to the equation of a damped harmonic oscillator. The characteristic exponents are

$$\lambda_{\pm} = -\frac{\Gamma}{4} \pm \sqrt{\left(\frac{\Gamma}{4}\right)^2 - V_{01}^2}. \tag{32}$$

Therefore the relevant regimes are determined by comparing V_{01} with $\Gamma/4$:

1. *Overdamped regime* ($V_{01} < \Gamma/4$): defining $\Omega_{\rm od} = \sqrt{(\Gamma/4)^2 - V_{01}^2}$,

$$\tilde{c}_0(t) = e^{-\Gamma t/4}\left[\cosh(\Omega_{\rm od}t) + \frac{\Gamma}{4\Omega_{\rm od}}\sinh(\Omega_{\rm od}t)\right], \tag{33}$$

$$\tilde{c}_1(t) = -i\frac{V_{01}}{\Omega_{\rm od}}e^{-\Gamma t/4}\sinh(\Omega_{\rm od}t). \tag{34}$$

   Population transfer from $|0\rangle$ is monotonic, and in the strong-damping limit $V_{01} \ll \Gamma$ the slow decay rate reduces to $\gamma_{\rm eff} \approx 2V_{01}^2/\Gamma$.

2. *Critical damping* ($V_{01} = \Gamma/4$):

$$\tilde{c}_0(t) = e^{-\Gamma t/4}\left(1 + \frac{\Gamma t}{4}\right), \qquad \tilde{c}_1(t) = -i\frac{\Gamma t}{4}e^{-\Gamma t/4}. \tag{35}$$

   This is the crossover between monotonic relaxation and oscillatory population exchange.

3. *Underdamped regime* ($V_{01} > \Gamma/4$): defining $\Omega_{\rm u} = \sqrt{V_{01}^2 - (\Gamma/4)^2}$,

$$\tilde{c}_0(t) = e^{-\Gamma t/4}\left[\cos(\Omega_{\rm u}t) + \frac{\Gamma}{4\Omega_{\rm u}}\sin(\Omega_{\rm u}t)\right], \tag{36}$$

$$\tilde{c}_1(t) = -i\frac{V_{01}}{\Omega_{\rm u}}e^{-\Gamma t/4}\sin(\Omega_{\rm u}t). \tag{37}$$

   The subsystem $\{|0\rangle, |1\rangle\}$ undergoes damped Rabi oscillations with oscillation frequency $2\Omega_{\rm u}$ and envelope $e^{-\Gamma t/4}$.

These three regimes provide the analytical interpretation of the Model B numerics discussed below: the bath sets the damping scale Γ, while V_{01} controls whether the upstream-to-doorway transfer is monotonic, critically damped, or oscillatory. Physically, the bath provides an effective irreversible decay channel for the doorway state $|1\rangle$ and therefore acts as a dissipative environment for the $\{|0\rangle, |1\rangle\}$ subsystem. The competition between the coherent coupling V_{01} and the dissipative rate Γ is what generates the three damping regimes.

III. RESULTS AND DISCUSSION

A. Model A: exponential decay and Fermi's golden rule

1. *Baseline dynamics*

A representative set of baseline parameters satisfying the hierarchy $\Delta E \gg V_{1k} \gg \rho^{-1}$ is

$$N = 200, \quad \Delta E = 10, \quad V_{1k} = 0.05, \quad E_1 = 0. \tag{38}$$

For these values $\rho = N/\Delta E = 20$, $\rho^{-1} = 0.05$, and the FGR decay rate is

$$\Gamma = 2\pi(0.05)^2 \times 20 \approx 0.314, \tag{39}$$

so that $\rho^{-1} \approx \Gamma/6.3 \ll \Gamma \ll \Delta E$, firmly inside the wide-band regime. Figure 2 shows $P_1(t)$ on both linear and logarithmic scales compared with the FGR benchmark $e^{-\Gamma t}$. The two curves are nearly indistinguishable throughout the observation window $t \in [0, 20]$, which spans more than six decay lengths ($5/\Gamma \approx 15.9$), confirming that the FGR prediction is quantitatively valid when the parameter hierarchy is satisfied.

2. *Effect of N*

To isolate the role of the bath discreteness from a change in the decay rate, we define the *reorganization energy*

$$\Lambda \equiv \sum_{k=1}^{N} |V_{1k}|^2. \tag{40}$$

In the terminology of the present finite-level model, Λ is simply the total squared doorway–bath coupling strength. It should not be confused with the reorganization energy of shifted harmonic-oscillator models, where the same phrase refers to an energy associated with equilibrium coordinate displacements of explicit vibrational modes. For the uniform-coupling case used here, $V_{1k} = V_N$ for all k, so

$$\Lambda = NV_N^2. \tag{41}$$

We then rescale the coupling as $V_N = V_{\rm ref}\sqrt{N_{\rm ref}/N}$ so that

$$\Lambda = NV_N^2 = N_{\rm ref}V_{\rm ref}^2 = \text{const.} \tag{42}$$

At fixed bandwidth ΔE, this also fixes the FGR rate because $\rho_N = N/\Delta E$ and therefore

$$\Gamma = 2\pi V_N^2 \rho_N = \frac{2\pi\Lambda}{\Delta E}. \tag{43}$$

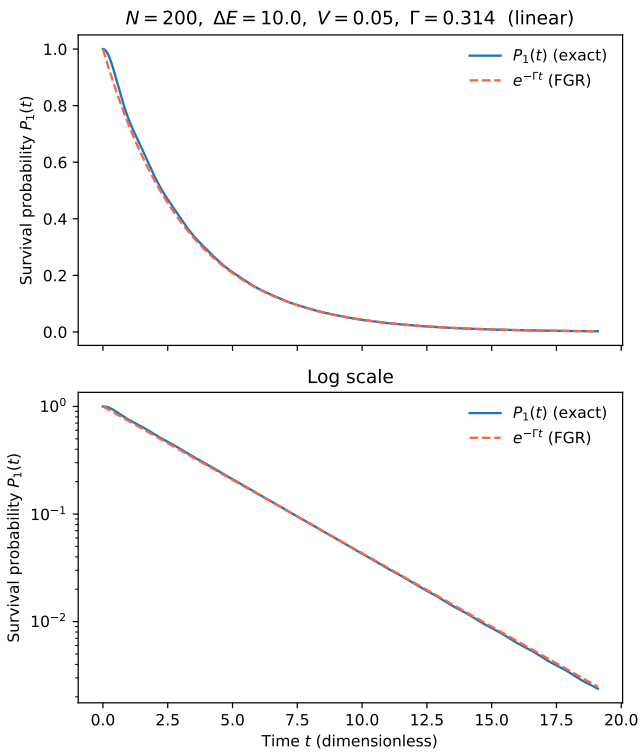

FIG. 2. Baseline dynamics of Model A with $N = 200$, $\Delta E = 10$, $V_{1k} = 0.05$, $E_1 = 0$ (dimensionless units, $\hbar = 1$). The density of states is $\rho = 20$, giving a FGR decay rate $\Gamma = 2\pi V_{1k}^2 \rho \approx 0.314$. Top: survival probability $P_1(t)$ (solid blue) versus the FGR prediction $e^{-\Gamma t}$ (dashed red) on a linear scale. Bottom: same data on a logarithmic scale, highlighting agreement over more than six orders of magnitude. The exact and FGR curves are virtually indistinguishable throughout the window $t \in [0, 20]$, demonstrating quantitative validity of the wide-band approximation when $\rho^{-1} \ll \Gamma \ll \Delta E$.

Thus varying N at fixed Λ changes only the bath discreteness and the recurrence time, not the continuum-limit decay rate. For the present parameters, $\Lambda = 0.5$ and $\Gamma \approx 0.314$ for all N. The Poincaré recurrence time

$$T_{\rm rec} = \frac{2\pi}{\Delta\omega} = \frac{2\pi(N-1)}{\Delta E}, \tag{44}$$

where $\Delta\omega = \Delta E/(N-1)$ is the uniform level spacing, grows linearly with N at fixed ΔE.

Figure 3 compares $P_1(t)$ for $N = 50$, 100, and 200 (with $V_N = 0.100$, 0.071, and 0.050, respectively) over a time window extending beyond the longest $T_{\rm rec}$. For all three cases the initial decay follows the common FGR exponential $e^{-\Gamma t}$ closely, but a sharp Poincaré revival restores substantial population to $|1\rangle$ at $t \approx T_{\rm rec}$ (annotated with arrows in the figure). The revival is most prominent for small N: at $N = 50$ ($T_{\rm rec} \approx 30.8$) the peak revival amplitude exceeds 0.8, while at $N = 200$ ($T_{\rm rec} \approx 125.0$) the revival occurs far outside the experimentally relevant time window and the exponential approximation is valid over many more decay lengths. On the logarithmic scale the depth of the trough before the revival deepens with increasing N, reflecting the improved continuum-limit approximation.

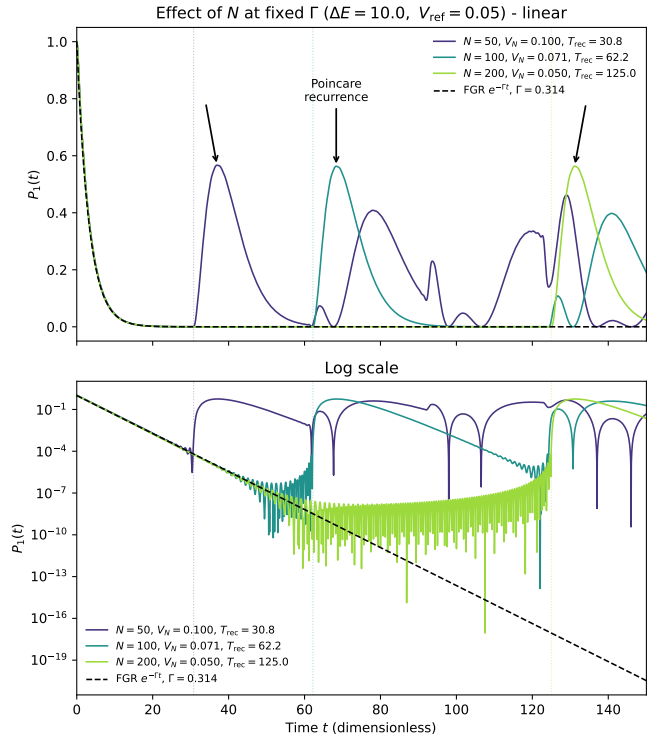

FIG. 3. Effect of the number of bath levels N on the survival probability $P_1(t)$ of Model A. The coupling is rescaled as $V_N = V_{\rm ref}\sqrt{N_{\rm ref}/N}$ with $V_{\rm ref} = 0.05$ and $N_{\rm ref} = 200$, keeping the reorganization energy $\Lambda = NV_N^2 = 0.5$ and hence the FGR rate fixed at $\Gamma \approx 0.314$ for all curves. Parameters: $\Delta E = 10$, $E_1 = 0$. Three bath sizes are shown: $N = 50$ ($V_N = 0.100$, $T_{\rm rec} \approx 30.8$, dark blue), $N = 100$ ($V_N = 0.071$, $T_{\rm rec} \approx 62.2$, teal), and $N = 200$ ($V_N = 0.050$, $T_{\rm rec} \approx 125.0$, yellow-green). The black dashed line is the FGR prediction $e^{-\Gamma t}$. Dotted vertical lines and arrows mark the Poincaré recurrence time $T_{\rm rec}$ for each N. Top: linear scale; bottom: logarithmic scale. All curves decay exponentially at rate Γ before the first revival, demonstrating that exponential decay is an intermediate-time approximation valid only for $t \ll T_{\rm rec}$.

3. *Effect of ΔE*

At fixed $N = 200$ and $V_{1k} = 0.05$, varying ΔE simultaneously changes both the density of states $\rho = N/\Delta E$ and the ratio $\Gamma/\Delta E = 2\pi V_{1k}^2 N/\Delta E^2$, which controls how well the wide-band approximation is satisfied. Figure 4 shows $P_1(t)$ for four bandwidths $\Delta E \in \{2, 5, 10, 20\}$, corresponding to FGR rates $\Gamma \approx \{1.57, 0.63, 0.31, 0.16\}$ and ratios $\Gamma/\Delta E \approx \{0.79, 0.13, 0.03, 0.008\}$.

For the two widest bandwidths ($\Delta E = 10$ and 20) the ratio $\Gamma/\Delta E \ll 1$ and the dynamics are cleanly exponential, with $P_1(t)$ indistinguishable from the respective FGR prediction. As ΔE narrows, $\Gamma/\Delta E$ grows and

the wide-band approximation begins to fail: strong oscillations are already evident at $\Delta E = 2$, the narrowest bandwidth shown, where $\Gamma/\Delta E \approx 0.79$ and the doorway state is strongly influenced by the nearby band edges. This onset of non-Markovian behavior is characterized by pronounced oscillations throughout the observation window. Conversely, larger ΔE pushes the band edges further from E_1 and deepens the Markovian limit, suppressing both oscillations and revivals.

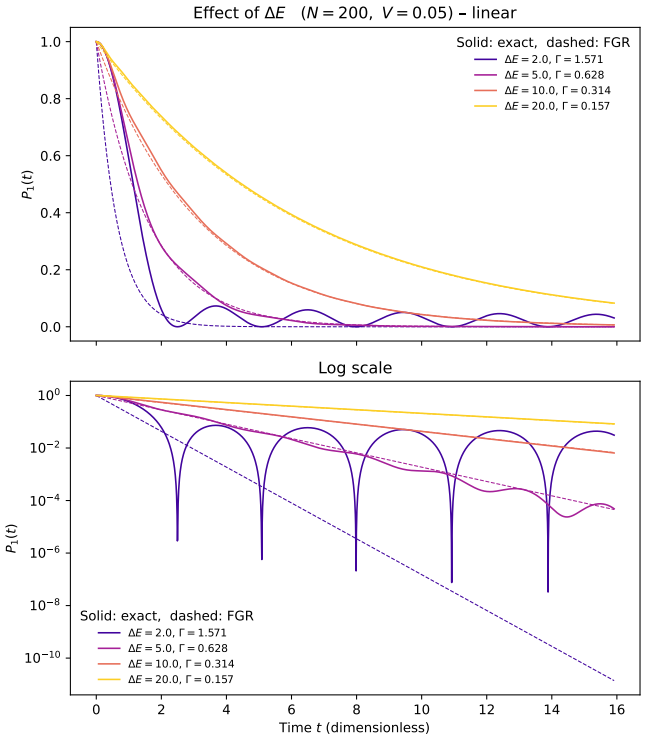

FIG. 4. Effect of the bath bandwidth ΔE on the survival probability $P_1(t)$ of Model A with $N = 200$, $V_{1k} = 0.05$, $E_1 = 0$. Four values are shown: $\Delta E = 2$ (magenta, $\Gamma \approx 1.57$), $\Delta E = 5$ (blue, $\Gamma \approx 0.63$), $\Delta E = 10$ (red, $\Gamma \approx 0.31$), $\Delta E = 20$ (yellow, $\Gamma \approx 0.16$). The black dashed line is the FGR prediction for $\Delta E = 10$. Top: linear scale; bottom: logarithmic scale. The narrowest bandwidth shown ($\Delta E = 2$, $\Gamma/\Delta E \approx 0.8$) already violates the wide-band approximation and exhibits strongly non-exponential, oscillatory dynamics. Wider bandwidths ($\Delta E \geq 10$, $\Gamma/\Delta E \leq 0.03$) satisfy the approximation and recover clean exponential decay.

4. *Effect of V_{1k}*

Increasing V_{1k} at fixed $N = 200$ and $\Delta E = 10$ raises Γ through $\Gamma = 2\pi V_{1k}^2 \rho$, accelerating the decay while also widening the resonance relative to the bath bandwidth. Figure 5 compares three coupling strengths $V_{1k} \in \{0.05, 0.1, 0.15\}$, corresponding to $\Gamma \approx \{0.314, 1.257, 2.827\}$ and $\Gamma/\Delta E \approx \{0.03, 0.13, 0.28\}$.

For the weakest coupling ($V_{1k} = 0.05$, $\Gamma/\Delta E \approx 0.03$) the exact curve tracks the FGR exponential essentially perfectly over the entire window. At $V_{1k} = 0.1$ ($\Gamma/\Delta E \approx 0.13$) the initial decay is slightly faster than predicted by FGR and mild oscillations appear at later times, signaling incipient breakdown of the wide-band approximation. At $V_{1k} = 0.15$ ($\Gamma/\Delta E \approx 0.28$) the departure is substantial: the exact decay is markedly faster than $e^{-\Gamma t}$ at early times and persistent Rabi-like oscillations dominate the long-time dynamics, reflecting coherent coupling to the collective band mode. These results confirm that the validity of FGR requires $\Gamma \ll \Delta E$, not merely $\Gamma < \Delta E$.

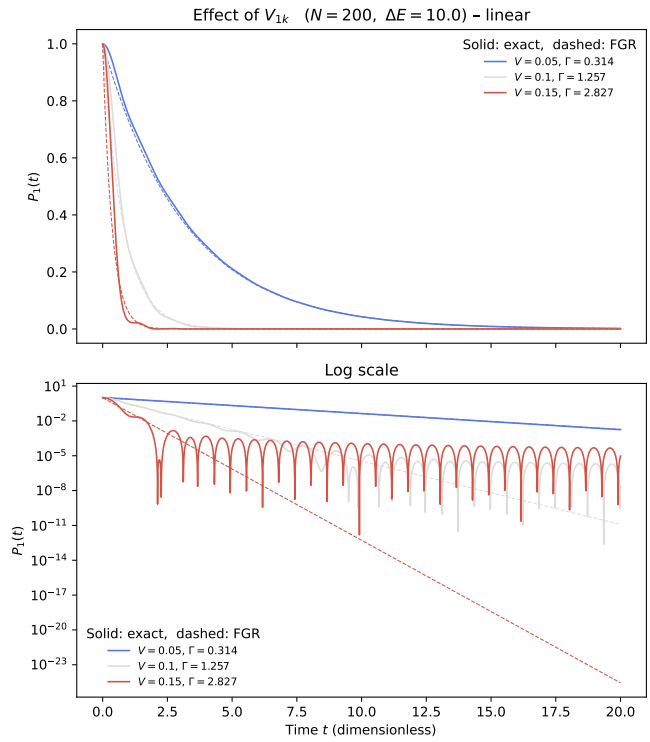

FIG. 5. Effect of the coupling strength V_{1k} on the survival probability $P_1(t)$ of Model A with $N = 200$, $\Delta E = 10$, $E_1 = 0$. Three values are shown: $V_{1k} = 0.05$ (blue, $\Gamma \approx 0.314$, $\Gamma/\Delta E \approx 0.03$), $V_{1k} = 0.10$ (silver/gray, $\Gamma \approx 1.257$, $\Gamma/\Delta E \approx 0.13$), and $V_{1k} = 0.15$ (red, $\Gamma \approx 2.827$, $\Gamma/\Delta E \approx 0.28$). Solid lines are exact TDSE results; dashed lines of matching colour are the corresponding FGR predictions $e^{-\Gamma t}$. Top: linear scale; bottom: logarithmic scale, $t \in [0, 20]$. As V_{1k} increases the resonance width Γ approaches the bandwidth ΔE, causing the exact decay to deviate progressively from the FGR exponential and eventually to exhibit sustained oscillations (crossover from Markovian-looking decay to coherent oscillatory dynamics).

5. *Uniform versus random-sign couplings*

For Model A, replacing uniform couplings $V_{1k} = V$ with equal-magnitude random-sign couplings $V_{1k} = \pm V$ leaves the survival probability $P_1(t)$ *unchanged*. This is a structural consequence of the Hamiltonian topology: since each manifold state $|k\rangle$ couples only to the single

doorway state $|1\rangle$, the sign of V_{1k} can be absorbed into a redefinition $|k\rangle \to -|k\rangle$, which is a valid unitary transformation that leaves all physical observables invariant. Gaussian couplings $V_{1k} \overset{\text{i.i.d.}}{\sim} \mathcal{N}(0, V^2)$ introduce seed-to-seed fluctuations, widening the envelope of $P_1(t)$ while preserving the mean decay rate.

Figure 6 shows $P_1(t)$ for all three coupling schemes ($N = 200$, $\Delta E = 10$, $V_{\text{rms}} = 0.05$, averaged over 1000 disorder realisations with $\pm 1\sigma$ shaded bands). As expected, the uniform and random-sign curves are numerically identical. The Gaussian ensemble mean also tracks the FGR prediction closely, but with a visible spread reflecting magnitude disorder: realisations with larger $\langle V_{1k}^2 \rangle^{1/2}$ decay faster while those with smaller values decay more slowly, smearing the revival structure. All three schemes satisfy $\langle \Gamma \rangle = 2\pi V_{\text{rms}}^2 \rho$ to within statistical uncertainty.

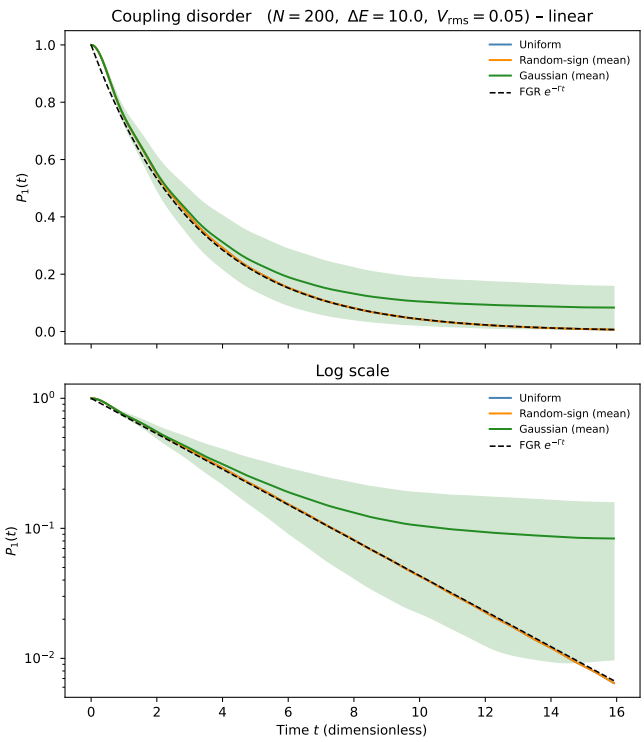

FIG. 6. Effect of coupling disorder on $P_1(t)$ in Model A ($N = 200$, $\Delta E = 10$, $V_{\text{rms}} = 0.05$, $E_1 = 0$). Three coupling schemes are compared: uniform $V_{1k} = V$ (blue solid), random-sign $V_{1k} = \pm V$ (orange, mean over 1000 realisations), and Gaussian $V_{1k} \overset{\text{i.i.d.}}{\sim} \mathcal{N}(0, V^2)$ (green, mean over 1000 realisations). Shaded bands denote ± 1 standard deviation across realisations. The black dot-dashed line is the FGR prediction $e^{-\Gamma t}$ with $\Gamma = 2\pi V^2 \rho \approx 0.314$. Top: linear scale; bottom: logarithmic scale. Uniform and random-sign results are numerically identical due to a gauge equivalence (sign redefinition of bath states). The Gaussian ensemble reproduces the same mean decay rate but exhibits broader realization-to-realization variance, especially at late times.

B. Model B: two-step relaxation dynamics

1. *Sequential population flow*

In Model B the population must flow from $|0\rangle$ to $|1\rangle$ before it can relax into the bath manifold. As derived in Sec. II C, integrating out the bath in the wide-band limit reduces the $|0\rangle$–$|1\rangle$ subspace to a damped two-level problem; on resonance, the amplitude $c_0(t)$ satisfies the same second-order equation as a damped harmonic oscillator [Eq. (31)], with the bath setting Γ and V_{01} playing the role of the restoring frequency. The critical threshold is $V_{01} = \Gamma/4 \approx 0.0785$. Figure 7 surveys five representative values of V_{01} at fixed $N = 200$, $\Delta E = 10$, $V_{1k} = 0.05$ ($\Gamma \approx 0.314$), $E_0 = E_1 = 0$, and $t_{\max} = 40$, spanning the full range from very overdamped to strongly underdamped dynamics.

Panels A and B show two overdamped cases. At $V_{01} = 0.01$ (panel A, $V_{01}/(\Gamma/4) = 0.13$), the system is deeply overdamped: the effective rate $\gamma_{\text{eff}} = 2V_{01}^2/\Gamma \approx 6.4 \times 10^{-4}$ is so small that P_0 remains near unity over the entire window, P_1 is a barely visible transient, and P_K stays essentially zero. Increasing to $V_{01} = 0.04$ (panel B, $V_{01}/(\Gamma/4) = 0.51$) accelerates the decay to $\gamma_{\text{eff}} \approx 0.010$; P_0 falls noticeably and a small P_1 transient develops, but the dynamics remains monotonic.

Panel C shows critical damping ($V_{01} = \Gamma/4 \approx 0.0785$), the boundary between regimes. The characteristic exponents $\lambda_\pm$ are degenerate, giving the P_0 envelope $e^{-\Gamma t/4}(1 + \Gamma t/4)$. Population reaches the bath as quickly as possible without oscillating, and the P_1 transient is larger than in either overdamped panel.

Panels D and E exhibit underdamped coherent dynamics. At $V_{01} = 0.20$ (panel D, $V_{01}/(\Gamma/4) = 2.55$), damped Rabi oscillations with frequency $2\Omega_{\text{u}} = 2\sqrt{V_{01}^2 - (\Gamma/4)^2} \approx 0.37$ are clearly resolved; P_K rises in a staircase pattern that mirrors each half-cycle of population transfer. At $V_{01} = 0.50$ (panel E, $V_{01}/(\Gamma/4) = 6.37$), the oscillation frequency $2\Omega_{\text{u}} \approx 0.99$ is large enough that roughly six full Rabi cycles are completed before the $e^{-\Gamma t/4}$ envelope damps the amplitude to zero, with P_K filling entirely within the window. In all five panels the WBA analytical prediction (dashed) agrees quantitatively with the exact TDSE result, validating the wide-band approximation across the full parameter range.

2. *Effect of $E_0 - E_1$*

The energy offset $\Delta_{01} = E_0 - E_1$ controls how efficiently population can flow from $|0\rangle$ to $|1\rangle$. Using the overdamped parameter $V_{01} = 0.04$ (panel B of Fig. 7), Fig. 8 scans $\Delta_{01} \in \{0, \pm 0.5, \pm 1.5, \pm 3.0\}$ with $t_{\max} = 100$, long enough to resolve the slow resonant decay $\gamma_0 = 4V_{01}^2/\Gamma \approx 0.020$.

At $\Delta_{01} = 0$ (resonance), P_0 decays at the maximum rate, P_1 builds up to its largest transient (panel A), and

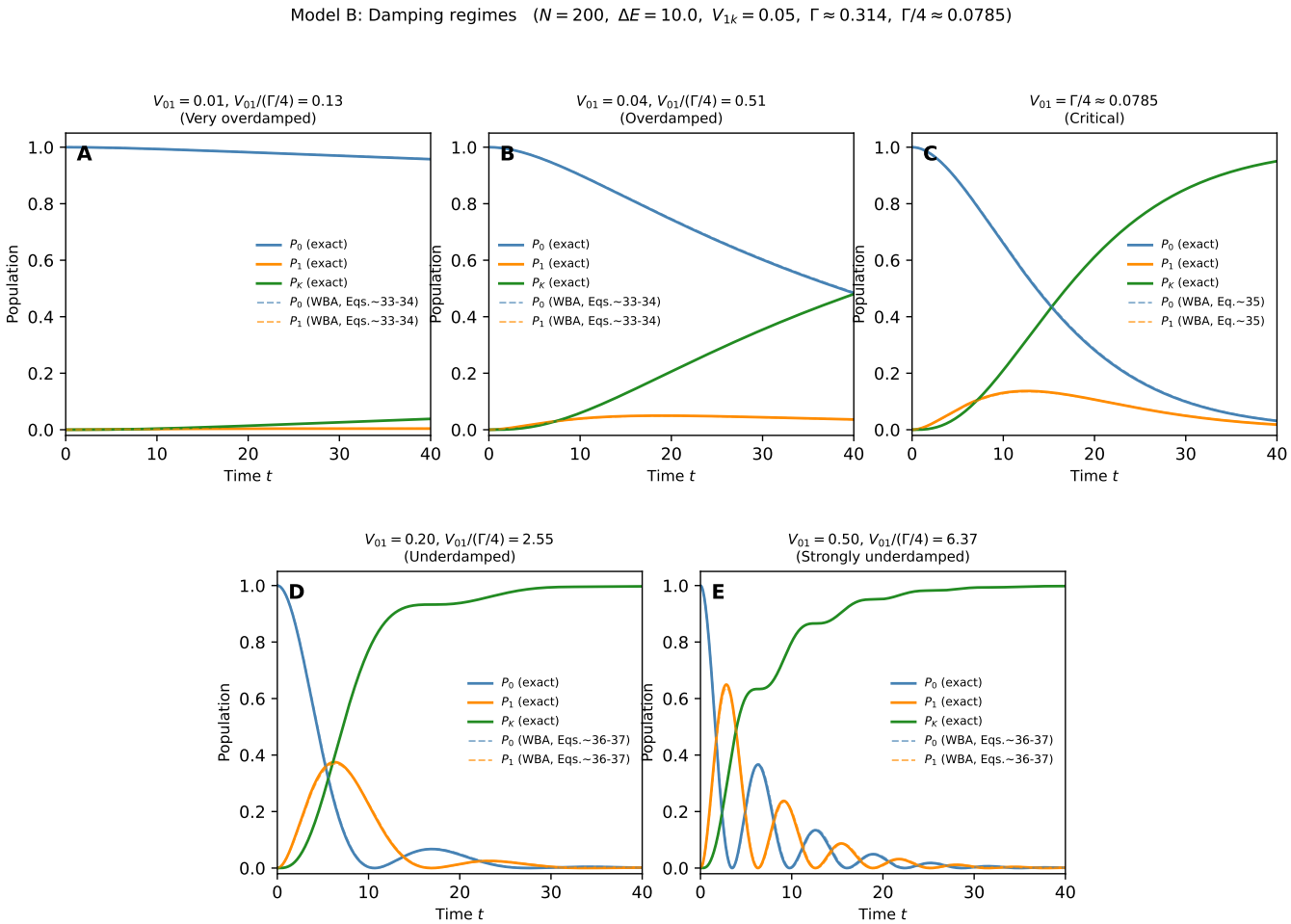

FIG. 7. **Five damping regimes of Model B**, classified by the ratio $V_{01}/(\Gamma/4)$ (Sec. II C). Parameters: $N = 200$, $\Delta E = 10$, $V_{1k} = 0.05$, $E_0 = E_1 = 0$, $\Gamma \approx 0.314$, $\Gamma/4 \approx 0.0785$; $t_{\max} = 40$. Solid lines: exact TDSE. Dashed lines: WBA analytical solution from Sec. II C, based on Eqs. 31 and 32. Blue: $P_0(t)$; orange: $P_1(t)$; green: $P_K(t) = 1 - P_0 - P_1$. Upper row (A–C): overdamped side of the transition. **(A)** $V_{01} = 0.01$ ($V_{01}/(\Gamma/4) = 0.13$, very overdamped): P_0 barely decays over the window; $\gamma_{\text{eff}} = 2V_{01}^2/\Gamma \approx 6.4 \times 10^{-4}$. **(B)** $V_{01} = 0.04$ ($V_{01}/(\Gamma/4) = 0.51$, overdamped): monotonic decay with $\gamma_{\text{eff}} \approx 0.010$. **(C)** $V_{01} = \Gamma/4 \approx 0.0785$ (critical): fastest non-oscillatory transfer; P_1 transient is maximal. Lower row (D–E): underdamped side. **(D)** $V_{01} = 0.20$ ($V_{01}/(\Gamma/4) = 2.55$): several damped Rabi cycles with $2\Omega_{\text{u}} \approx 0.37$. **(E)** $V_{01} = 0.50$ ($V_{01}/(\Gamma/4) = 6.37$): rapid coherent oscillations ($2\Omega_{\text{u}} \approx 0.99$, period ≈ 6.3) with full damping envelope $e^{-\Gamma t/4}$ within the window. WBA dashed lines agree with the exact TDSE in all panels.

P_K fills up rapidly to near unity (panel B). As $|\Delta_{01}|$ increases, the effective transfer rate drops sharply: at $|\Delta_{01}| = 0.5$ it falls to $\approx 9\%$ of the resonant value, and at $|\Delta_{01}| = 3.0$ to below 0.3%. This rapid suppression is captured by the Lorentzian (panel C), whose half-width at half-maximum is the bath-induced linewidth $\Gamma/2 \approx 0.157$, far smaller than the bath bandwidth $\Delta E/2 = 5$.

A key observation is that positive and negative detunings produce *identical* $P_0(t)$ and $P_1(t)$ curves (solid and dashed lines overlap exactly in panels A and B). This follows from the energy-reflection symmetry of the model: the bath spectrum is symmetric about $E_1 = 0$, so $\Delta_{01} \to -\Delta_{01}$ is equivalent to reversing all energies, which leaves populations unchanged.

The effective decay rate plotted in panel C is extracted from the TDSE data as follows. In the overdamped regime the fast normal mode of H_{eff} decays on the timescale $\sim 1/\Gamma \approx 3$, leaving a single slow exponential $P_0(t) \approx A\, e^{-\gamma_{\text{eff}} t}$ for $t \gg 1/\Gamma$. We therefore fit $\ln P_0(t)$ to a straight line over the second half of the simulation window ($t \in [50, 100]$), where the fast transient has fully died out and P_0 is still well above numerical noise. The slope of the linear fit gives γ_{eff} directly. This procedure is reliable provided P_0 has not yet been fully depleted, which is satisfied for all detunings shown ($P_0(100) \gtrsim 10^{-2}$ even at resonance).

Panel C demonstrates that the WBA Lorentzian prediction agrees quantitatively with the numerically extracted rates across the full scan, confirming that the Breit–Wigner line shape governs the resonance structure

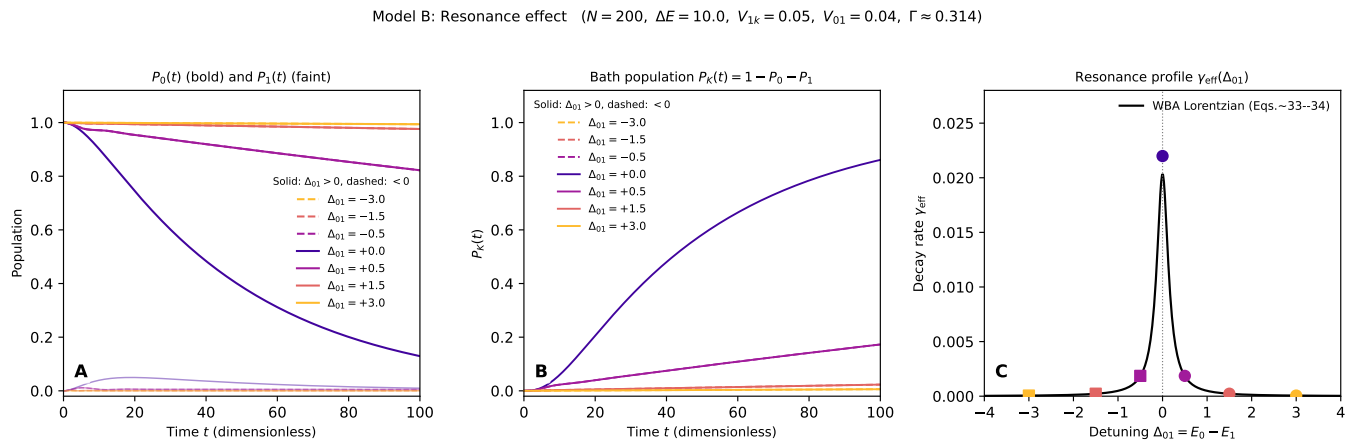

FIG. 8. **Resonance effect in Model B.** Parameters: $N = 200$, $\Delta E = 10$, $V_{1k} = 0.05$, $V_{01} = 0.04$ (overdamped, panel B of Fig. 7), $\Gamma \approx 0.314$; $t_{\max} = 100$. Detunings $\Delta_{01} \in \{0, \pm 0.5, \pm 1.5, \pm 3.0\}$; solid lines: $\Delta_{01} > 0$, dashed: $\Delta_{01} < 0$; same color for $\pm|\Delta_{01}|$. **(A)** Both $P_0(t)$ (bold) and $P_1(t)$ (faint, same color): P_0 decays fastest at resonance ($\Delta_{01} = 0$) while P_1 builds up to its largest transient; both are progressively suppressed as $|\Delta_{01}|$ grows. **(B)** Bath population $P_K(t) = 1 - P_0 - P_1$: P_K fills up rapidly at resonance and saturates near unity, while off-resonance population remains trapped in $|0\rangle$ over the window. Positive and negative detunings give identical dynamics, confirming the energy-reflection symmetry of the bath. **(C)** Extracted population decay rate $\gamma_{\rm eff}$ versus Δ_{01} (circles: $\Delta_{01} > 0$; squares: $\Delta_{01} < 0$), overlaid with the WBA Lorentzian $\gamma_{\rm eff} = V_{01}^2 \Gamma / (\Delta_{01}^2 + (\Gamma/2)^2)$ (black curve). The Lorentzian has width $\Gamma/2 \approx 0.157$ and peak $4V_{01}^2/\Gamma \approx 0.020$.

of the two-step decay process.

3. *Effect of N in Model B*

Figure 9 compares $N = 50$, 100, 200 with V_{1k} scaled to keep $\Gamma \approx 0.314$ and $\Lambda = NV_{1k}^2 = 0.5$ fixed, and with $V_{01} = 0.04$ (overdamped, same parameter regime as Fig. 8). The black dashed line shows the $N \to \infty$ WBA analytical result from Sec. II C: in this limit the bath is a true continuum and no Poincaré revival occurs, so P_0, P_1, P_K vary smoothly for all time. The three $P_0(t)$ curves in panel A are nearly superimposed and closely track the WBA prediction, confirming that the rate-limiting step is the $|0\rangle \to |1\rangle$ coupling, which depends only on V_{01} and is independent of the bath density of states. Panel B shows that the doorway state $P_1(t)$ carries a small transient determined by V_{01} and Γ; small revivals appear at each $T_{\rm rec}$ as population briefly returns from the bath to $|1\rangle$. Panel C confirms that P_K fills identically for all N on the initial decay, with brief dips at $T_{\rm rec}$ corresponding to the Poincaré recurrence: the discrete bath returns its population to the system before it re-enters the bath manifold. The revival amplitude is smaller than in Model A (Fig. 3) because population must traverse the extra $|1\rangle \to |0\rangle$ step to reach P_0, attenuating the recurrence signal. Larger N pushes $T_{\rm rec}$ to longer times, extending the window over which the Markovian exponential approximation is valid.

4. *Coupling disorder in Model B*

Figure 10 overlays three coupling schemes for Model B ($N = 200$, $\Delta E = 10$, $V_{1k,\rm rms} = V_{01} = 0.05$) across three panels showing $P_0(t)$ (panel A), $P_1(t)$ (panel B), and the bath population $P_K(t) = 1 - P_0 - P_1$ (panel C).

Random-sign V_{1k} is again indistinguishable from the uniform case. As in Model A, flipping the sign of an individual coupling $V_{1k} \to -V_{1k}$ is a gauge transformation on bath state $|k\rangle$ that leaves all populations unchanged, so the mean and the ensemble spread both collapse onto the uniform baseline in all three panels.

Gaussian V_{1k} introduces genuine disorder. Averaged over 1000 realisations, $P_0(t)$ acquires a moderate but well-converged sample-to-sample variance because different realisations carry different effective Γ; the variance grows over the observation window. $P_1(t)$ shows a larger relative variance near its transient peak: realisations with stronger bath couplings drain $|1\rangle$ faster, producing an earlier and smaller peak. The bath population $P_K(t)$ mirrors the complementary variance: when P_0 and P_1 fluctuate, the excess probability must reside in the bath, so the spread in P_K reflects the combined disorder from both upstream states.

The sign of the upstream coupling V_{01} is a global gauge freedom. A sign change $V_{01} \to -V_{01}$ is equivalent to $|0\rangle \to -|0\rangle$, which shifts the phase of the initial amplitude without affecting any population $|\langle j|\psi(t)\rangle|^2$. Randomising the sign of V_{01} therefore produces dynamics identical to the uniform case, and no disorder effect is observable.

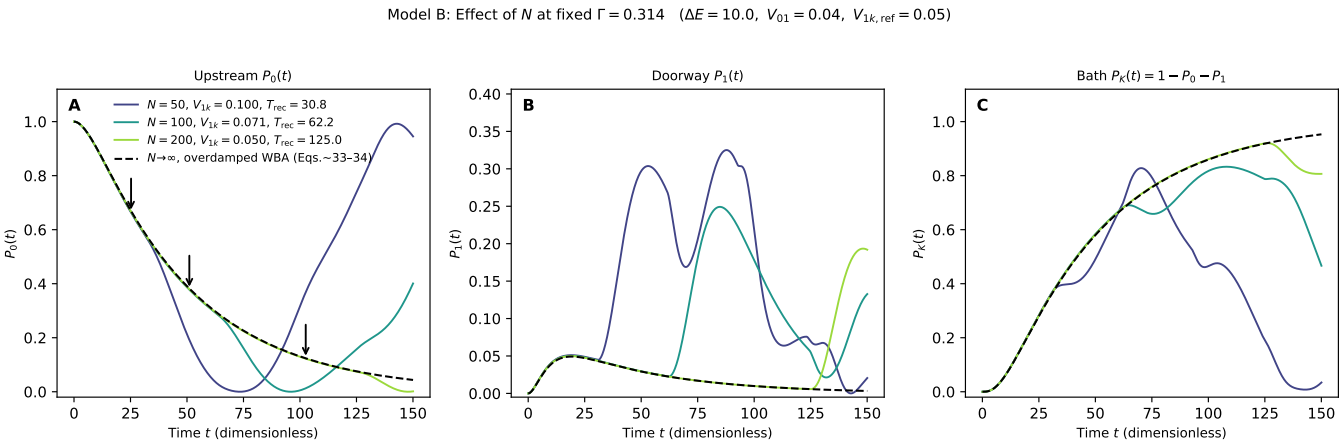

FIG. 9. **Effect of bath size N in Model B at fixed reorganization energy.** Following the design of Fig. 3, V_{1k} is scaled as $V_{1k}(N) = V_{\rm ref}\sqrt{N_{\rm ref}/N}$ with $V_{\rm ref} = 0.05$, $N_{\rm ref} = 200$ so that $\Lambda = NV_{1k}^2 = 0.5$ and $\Gamma \approx 0.314$ are held constant. The upstream coupling is fixed at $V_{01} = 0.04$ (overdamped, same as Fig. 8); $\Delta E = 10$, $E_0 = E_1 = 0$. Parameter values: $N = 50$ ($V_{1k} = 0.100$, $T_{\rm rec} = 30.8$), $N = 100$ ($V_{1k} = 0.071$, $T_{\rm rec} = 62.2$), $N = 200$ ($V_{1k} = 0.050$, $T_{\rm rec} = 125.0$). Black dashed line: $N \to \infty$ WBA analytical result from Sec. II C. Black downward arrows point to the Poincaré revival feature (local maximum in P_0, P_1; local minimum in P_K) for each N. **(A)** Upstream population $P_0(t)$: all three finite-N curves nearly coincide with the WBA limit, confirming that P_0 is rate-limited by V_{01}, not N. Revival bumps visible at each $T_{\rm rec}$ as population briefly returns from the bath. **(B)** Doorway population $P_1(t)$: revival peaks at $T_{\rm rec}$ grow in prominence as N decreases (shorter recurrence time). **(C)** Bath population $P_K(t) = 1 - P_0 - P_1$: all curves fill at the same rate; brief dips at $T_{\rm rec}$ coincide with the revivals in panels A and B.

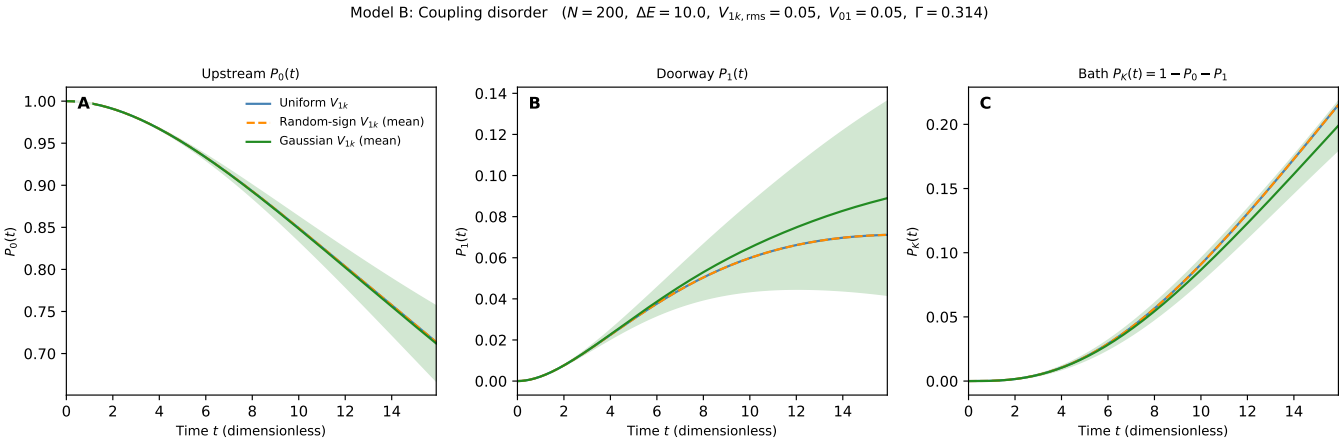

FIG. 10. **Effect of coupling disorder in Model B.** Parameters: $N = 200$, $\Delta E = 10$, $V_{1k,\rm rms} = 0.05$, $V_{01} = 0.05$, $\Gamma = 0.314$; ensemble of 1000 realisations for stochastic cases. Shaded bands show ± 1 standard deviation over realisations. Panel A: upstream population $P_0(t)$. Panel B: doorway population $P_1(t)$. Panel C: bath population $P_K(t) = 1 - P_0 - P_1$. *Blue (uniform V_{1k}):* deterministic baseline. *Orange dashed (random-sign V_{1k}):* mean coincides with the uniform curve and the band collapses to zero, confirming the gauge equivalence that extends from Model A to Model B when V_{01} is fixed. *Green (Gaussian V_{1k}):* genuine disorder shifts the mean and broadens $P_1(t)$ near its transient peak; the variance is reflected symmetrically in $P_K(t)$ while leaving the slower $P_0(t)$ decay comparatively unaffected.

IV. CONCLUSIONS

We have carried out a combined analytical and numerical study of quantum relaxation in two finite-dimensional doorway-state models. The principal conclusions are as follows.

1. **Model A: Green's-function theory and exponential decay.** For a single doorway state coupled to a bath manifold, the retarded Green's-function / self-energy treatment reproduces the Fermi's-golden-rule result $P_1(t) = e^{-\Gamma t}$ under the wide-band approximation. The exact TDSE numerics confirm that this Markovian exponential regime is realized when $\rho^{-1} \lesssim \Gamma \ll \Delta E$, and that the agreement improves as the bath becomes denser and

broader.

2. **Finite-size effects and breakdown of the wide-band regime.** In a finite bath the exponential law is only an intermediate-time approximation. The recurrence time $T_{\rm rec} \approx 2\pi N/\Delta E$ sets the onset of quantum Poincaré recurrences, while small ΔE, small N, or large V_{1k} drive the system out of the wide-band limit and generate non-Markovian oscillations. Conversely, if V_{1k} becomes too small compared with ρ^{-1}, the manifold no longer behaves as a continuum and the FGR picture also fails.

3. **Model B: damped two-level dynamics under the wide-band approximation.** When the bath coupled to $|1\rangle$ satisfies the same wide-band conditions, Model B reduces to an effective damped two-level system for the pair $\{|0\rangle, |1\rangle\}$ only because the doorway state is assumed, under the wide-band approximation, to decay Markovianly at rate Γ. The analytical theory then predicts three regimes separated by $V_{01} = \Gamma/4$: overdamped transfer for $V_{01} < \Gamma/4$, critical damping at $V_{01} = \Gamma/4$, and damped Rabi oscillations for $V_{01} > \Gamma/4$. The exact TDSE calculations agree quantitatively with this classification over the parameter range where the WBA is valid.

4. **Resonance structure in Model B.** The energy offset $\Delta_{01} = E_0 - E_1$ acts as a detuning parameter for the upstream-to-doorway transfer. In the overdamped regime the effective decay rate follows the Lorentzian form $\gamma_{\rm eff} = V_{01}^2\Gamma/(\Delta_{01}^2 + (\Gamma/2)^2)$, so population transfer is maximal at resonance and strongly suppressed for large $|\Delta_{01}|$. Positive and negative detunings give identical dynamics due to the energy-reflection symmetry of the model.

5. **Effect of coupling disorder.** Random-sign couplings are exactly gauge-equivalent to uniform couplings and therefore leave all populations unchanged in both models. By contrast, Gaussian-distributed bath couplings introduce genuine disorder: the mean dynamics remains close to the corresponding wide-band prediction, but sample-to-sample fluctuations broaden the envelopes and are most visible in the transient doorway population and in late-time recurrence features.

Taken together, these results provide a unified picture of doorway-state relaxation in finite systems. The wide-band approximation is the bridge from finite Hamiltonians to irreversible-looking decay, and the exact numerics show where that bridge holds and where it fails. More fundamentally, irreversible relaxation is not an intrinsic property of a finite quantum system, but an emergent continuum-limit phenomenon in which phase randomization suppresses coherent recurrences over the timescale of observation.

At the same time, the present study remains intentionally minimal: the bath is represented by discrete levels, the analysis is effectively restricted to zero-temperature relaxation, and no explicit molecular vibrational coordinates are included. These simplifications clarify the origin and breakdown of the wide-band approximation, but they also limit direct chemical realism.

Future work should embed the present doorway-state picture in more microscopic open-system models. In particular, replacing the discrete phenomenological bath by an explicit oscillator environment, such as a Caldeira–Leggett model [2, 4, 5], would allow temperature effects, spectral densities, memory kernels, and vibronic structure to be treated on the same footing. Even in their present form, however, these models provide a useful minimal framework for understanding relaxation in molecular systems, nanoscale devices, and open quantum systems with structured environments.

DATA AND CODE AVAILABILITY

The data and code that support the findings of this study are available at `https://github.com/Okita0512/CHEM-5260-Midterm-Project`.

ACKNOWLEDGMENTS

The author thanks Prof. Abraham Nitzan for designing this midterm project. The author also thanks the students of CHEM 5260 (Spring 2026) whose numerical experiments informed and inspired the analysis presented here. The author acknowledges Claude Code for assistance in generating the Python scripts used for data production and plotting, and ChatGPT for assistance with writing and revising this document.

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